

Budapest University of Technology and Economics

Faculty of Natural Sciences

M3 - Study of Comparators

Lab report

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Abstract

1 Theoretical introduction

2 Measurement tools

For this measurement, we used the following list of tools:

- ST μA741CN operational amplifier
- OK Industries Inc. CM-140 L breadboard
- Resistors: 22 kΩ (1 piece), 10 kΩ (2 pieces), 100 kΩ (1 piece)
- Jumping wires (6 pieces)
- Crocodile clips (6 pieces)
- Banana plugs (6 pieces)
- TI TL071CP operational amplifier
- Capacitor: 6.8 nF (1 piece)
- HM8040-2 HAMEG Triple power supply
- HM 8011-3 HAMEG Digital multimeter
- SIGLENT SDS 1072CML Digital storage oscilloscope
- Handheld multimeter

3 Measurement setup

4 Measurements

4.1 Switching voltage of the comparator with an operational amplifier

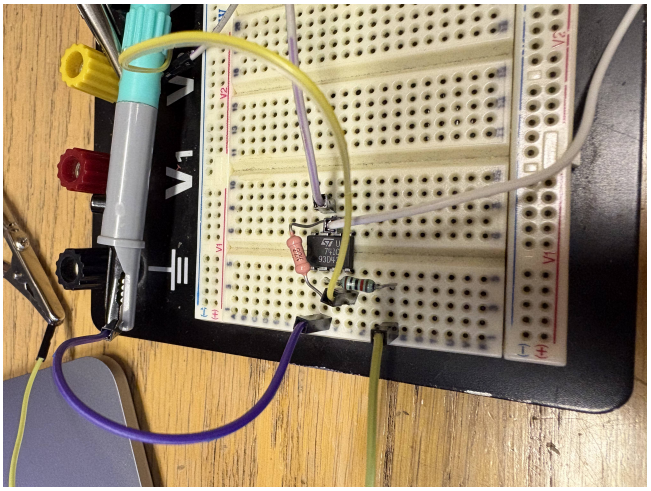


Figure 1: a

4.2 New switching voltage in Schmitt-trigger with operational amplifier

4.3 Waveforms of the relaxation oscillator with operational amplifier

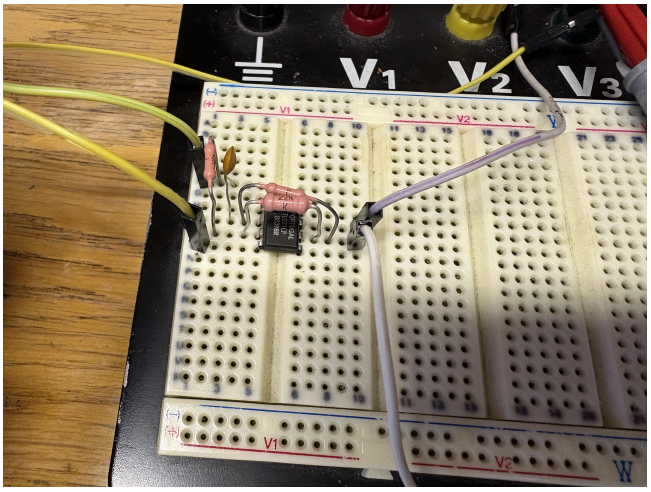
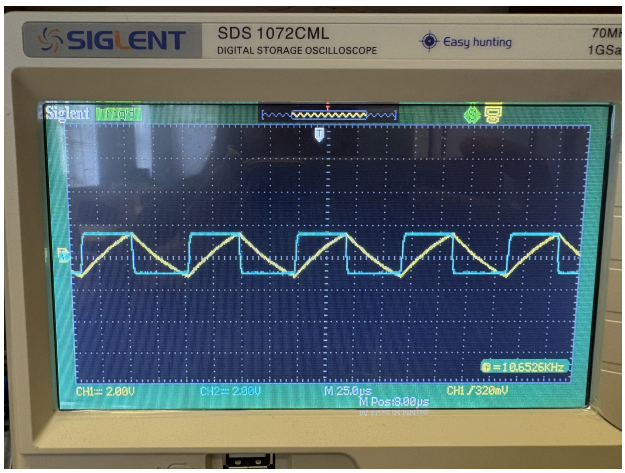
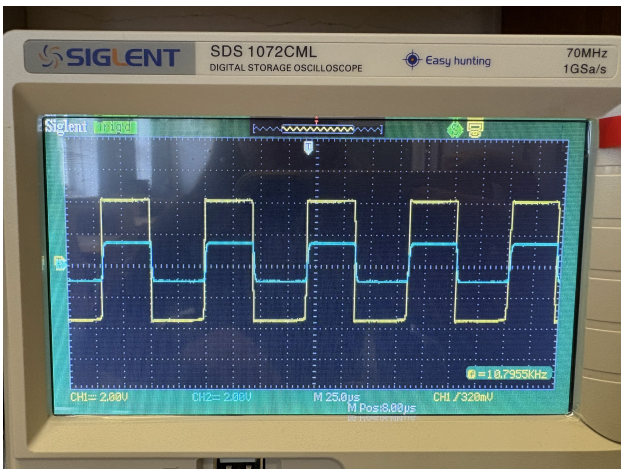


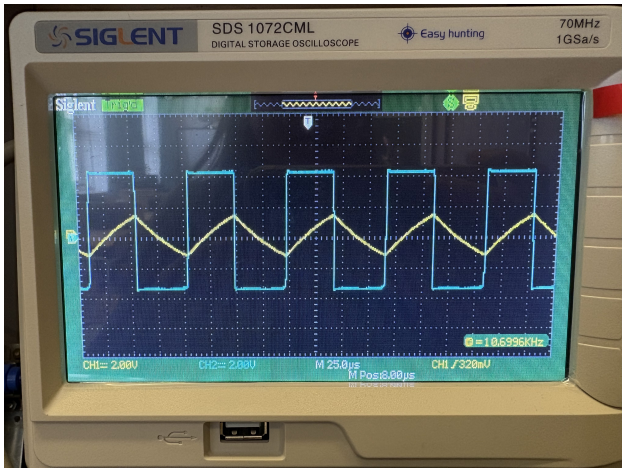
Figure 2: b



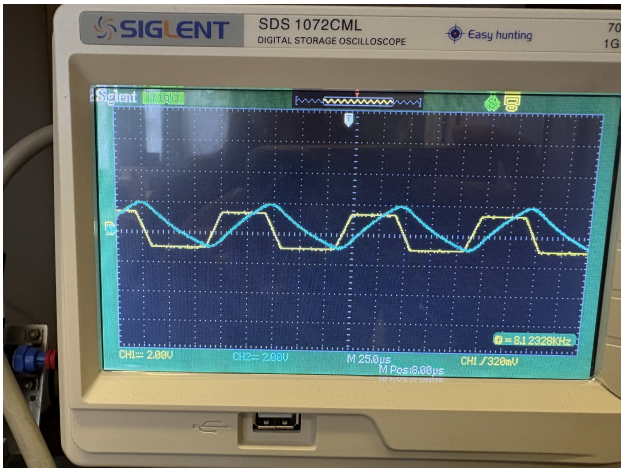
(a) c



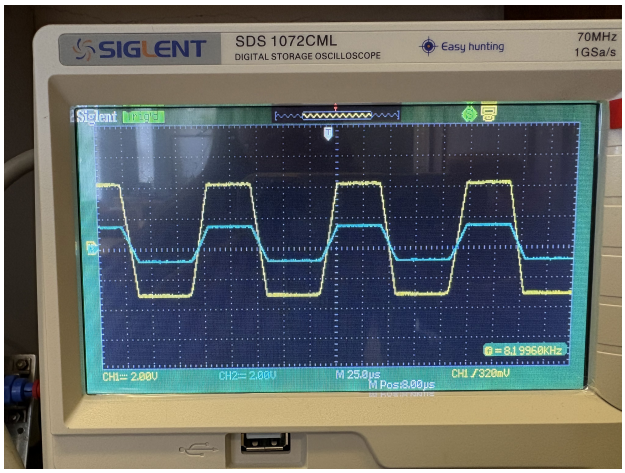
(b) d



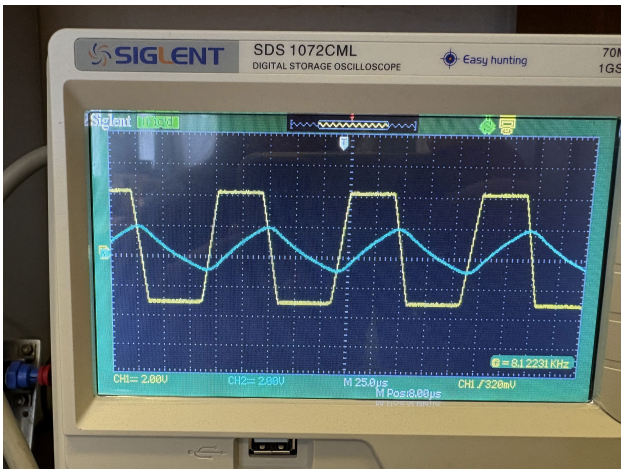
(c) e



(d) f



(e) g



(f) h

Figure 3: Oscilloscope measurements of the integrator circuit for sinusoidal, square and triangular input signals, with and without the additional resistor R_P.

4.4 Switching thresholds of the transistor Schmitt trigger circuit

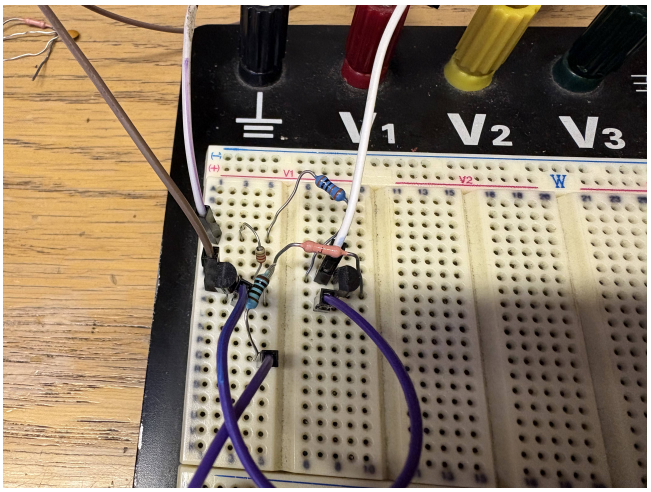


Figure 4: f

5 Conclusion